

Academic Literature Survey and Mathematical Synthesis: Bohmian Mechanics, Invariant Set Theory, and Asymptotic Safety in Unified Quantum Gravity (Version 6.3)

Executive Summary & Taxonomy Matrix

The primary challenge in modern theoretical physics remains the formulation of a self-consistent, non-perturbative quantum theory of gravity that simultaneously resolves the foundational problems of quantum measurement, non-locality, and ultraviolet divergences. Three independent theoretical frameworks address these challenges from distinct foundational viewpoints. Bohmian Mechanics (BM) reinterprets quantum phenomena through an ontic, deterministic trajectory formulation guided by a wave-functional¹. Invariant Set Theory

(IST) replaces continuous Hilbert spaces with a non-computable, $\mathbb{P}$ -adic fractal state-space geometry I_U , providing a deterministic, locally causal mechanism that evades Bell-type no-go theorems by violating measurement independence without fine-tuning³. Asymptotic Safety (AS) addresses quantum gravity through the Functional Renormalization Group (fRG), establishing a non-perturbative ultraviolet (UV) fixed point where coupling constants remain finite at arbitrarily high energy scales¹.

Mapping the intersections, mathematical bridges, and unresolved research gaps across these three theoretical pillars provides a foundation for synthesizing an advanced unified model, designated as Framework Version 6.3. Existing research is highly fragmented: Bohmian Mechanics and Asymptotic Safety intersect almost exclusively in comparative semiclassical gravity calculations and cosmological bounce dynamics¹. Invariant Set Theory and Bohmian Mechanics share a commitment to determinism and realism, yet Tim Palmer explicitly

differentiates IST's local, ψ -epistemic $\mathbb{P}$ -adic fractal state space from Bohmian mechanics'

non-local, ψ -ontic configuration-space guidance equations³. The intersection between Invariant Set Theory and Asymptotic Safety—specifically mapping scale-dependent spacetime spectral dimensions at UV fixed points to state-space fractal dimensions—remains completely unexplored, as does the full triadic domain across all three frameworks.

Domain Intersection	Primary Literature & Contributors	Core Formalisms & Mathematical Constructs	Maturity Level Classification
Bohmian Mechanics + Asymptotic Safety (BM + AS)	Bonanno & Reuter ⁶ ; Ali, Das, & Khalil ⁶ ; Pinto-Neto & Fabris ¹ ; Maes ²	Scale-dependent quantum potentials Q_k , Quantum Raychaudhuri Equations (QRE), RG-improved Schwarzschild metrics vs. Bohmian quantum-corrected metrics, fRG-derived field guidance equations.	Sparse Precedents
Invariant Set Theory + Bohmian Mechanics (IST + BM)	Palmer ³ ; Passon ⁸ ; Norsen ⁸	p -adic metric spaces versus Euclidean configuration spaces $\mathbb{R}^{3N}$, non-computable Cantor sets I_U , counterfactual setting restrictions vs. non-local action-at-a-distance.	Sparse Precedents
Invariant Set Theory + Asymptotic Safety (IST + AS)	Palmer ³ ; Reuter & Percacci ¹	Mapping scale-dependent spacetime spectral dimensions ($d_s \rightarrow 2$ at the UV fixed point) to	Completely Unexplored Territory (Gap)

		state-space fractal Hausdorff dimensions ($D_H < D_{\text{Euclidean}}$).	
Triadic Domain (BM + IST + AS)	Targeted for Framework Version 6.3 Synthesis	Scale-dependent pilot-wave field hydrodynamics evolving on non-computable $\mathcal{P}$ -adic fractal invariant manifolds driven by non-perturbative fRG fixed points.	Completely Unexplored Territory (Gap)

Detailed Literature Analysis by Domain Intersection

Domain A: Bohmian Mechanics + Asymptotic Safety (BM + AS)

The academic intersection between Bohmian Mechanics and Asymptotic Safety operates primarily through comparative semiclassical spacetime models, effective field theory formulations, and quantum cosmological bounce scenarios¹. In standard Asymptotic Safety, the Functional Renormalization Group (fRG) formulated via the Wetterich effective average action equation,

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)} + R_k \right)^{-1} \partial_k R_k \right]$$

yields scale-dependent running coupling constants, such as Newton's constant

$G(k) = \frac{G_0}{1 + \omega G_0 k^2}$ ⁶. Bonanno and Reuter applied this formalism to construct Renormalization Group (RG) improved Schwarzschild metrics, wherein the momentum cutoff k is identified with the inverse radial distance scale $k(r) \sim 1/r$, resulting in horizon modification and singularity softening at Planckian scales⁶.

In parallel, researchers in Bohmian quantum cosmology derived quantum-corrected metrics using the Quantum Raychaudhuri Equation (QRE) developed by Das, and later expanded by Ali and Khalil⁶. By replacing classical geodesic velocity fields with the Bohmian velocity field

$u_\mu = \frac{1}{m} \partial_\mu S$ (where S is the phase of the Klein-Gordon or Wheeler-DeWitt wavefunction $\Psi = Re^{iS/\hbar}$), the classical Raychaudhuri expansion factor acquires an explicit quantum potential stress tensor term⁶:

$$\frac{d\theta}{d\tau} = -\frac{1}{3}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu + \frac{\hbar^2}{m^2} \nabla_\mu \left(\frac{\square\sqrt{\rho}}{\sqrt{\rho}} \right) u^\mu$$

Comparative studies evaluating light deflection angles and black hole metric modifications reveal that both approaches independently predict similar physical corrections, such as the suppression of gravitational singularities and modifications to optical deflection angles⁶. However, their mathematical origins are fundamentally distinct: Asymptotic Safety attributes these corrections to the non-perturbative running of gravitational couplings toward a non-trivial UV fixed point (g_*, λ_*) , whereas Bohmian formulations attribute them to non-zero

quantum potential terms $Q = -\frac{\hbar^2}{2m} \frac{\square\sqrt{\rho}}{\sqrt{\rho}}$ acting on background fluid dynamics¹.

A direct synthesis of these two approaches has been initiated by Maes, who investigated the explicit renormalization and asymptotic safety of Bohmian field configurations². In this

framework, the scale-dependent guidance equation and running quantum potential $Q_k(x)$ are derived directly from the effective average action $\Gamma_k[\phi]$:

$$v_k^\mu(x) = \frac{1}{m} g_k^{\mu\nu}(x) \partial_\nu S_k(x), \quad Q_k(x) = -\frac{\hbar^2}{2m} \frac{\square_k \sqrt{\rho_k(x)}}{\sqrt{\rho_k(x)}}$$

Here, the probability density $\rho_k(x)$ and phase $S_k(x)$ evolve along the fRG trajectory from the infrared to the ultraviolet fixed point, providing a scale-covariant trajectory representation of quantum gravity². Furthermore, state-sum models on piecewise linear (PL) spacetimes have incorporated de Broglie-Bohm guidance equations alongside effective action formalisms, circumventing observer-dependent collapse while preserving non-perturbative finiteness analogous to Asymptotic Safety¹⁰.

Domain B: Invariant Set Theory + Bohmian Mechanics (IST + BM)

The relationship between Invariant Set Theory and Bohmian Mechanics centres on deterministic realism, local causality, and the ontology of the quantum state³. Both frameworks reject the orthodox Copenhagen interpretation, asserting that quantum probabilities reflect incomplete knowledge of underlying deterministic dynamics².

In Bohmian Mechanics, physical reality consists of point particles existing in 3D physical space,

guided deterministically through $3N$ -dimensional configuration space by an ontic pilot wave

$$\psi(q_1, \dots, q_N, t) \quad \text{2. Non-locality is explicit: the guidance equation} \quad v_i = \frac{\hbar}{m_i} \text{Im} \left(\frac{\nabla_i \psi}{\psi} \right)$$

implies instantaneous action-at-a-distance across spacelike separations, necessitating a preferred spatial foliation of spacetime that breaks manifest Lorentz covariance at the fundamental level².

Conversely, Tim Palmer's Invariant Set Theory maintains local causality in physical spacetime by redefining the global state space of the universe³. IST postulates that cosmological states

belong to a zero-measure fractal set I_U , dynamically invariant under universal deterministic evolution laws³. The geometry of I_U is homeomorphic to the algebraic set of p -adic integers $\mathbb{Z}_p$, equipped with a non-Archimedean p -adic metric:

$$d_p(x, y) = p^{-k}$$

where k is the highest power of p dividing the integer difference $(x - y)$ ³.

IST evades Bell's theorem, the Pusey-Barrett-Rudolph (PBR) theorem, and the Kochen-Specker

theorem by violating the Measurement Independence assumption $\rho(\lambda|a, b) = \rho(\lambda)$ ³.

Crucially, IST achieves this without fine-tuning or conspiracy: counterfactual state-space

trajectories corresponding to alternative measurement settings a' or b' simply do not reside

on the fractal invariant manifold I_U ³. Because states off I_U are physically undefined, counterfactual statements regarding alternative measurement outcomes are mathematically unrepresentable within the theory³. Standard quantum mechanics emerges from IST as a

singular limit as $p \rightarrow \infty$ ³. For any finite integer $p = 2^N + 1$, complex Hilbert space

operators provide an approximation of the underlying p -adic state-space dynamics³. While Bohmian Mechanics uses non-local configurations in continuous Euclidean space to reproduce quantum phenomena, IST attributes quantum phenomena to a local, non-computable fractal geometry in state space³.

Domain C: Invariant Set Theory + Asymptotic Safety (IST + AS)

The intersection between Invariant Set Theory and Asymptotic Safety constitutes a **Verified**

Research Gap. No published literature directly connects Palmer's p -adic state-space model with the Weinberg-Reuter fRG program. However, a significant mathematical bridge exists between their respective treatments of fractal geometry¹.

In Asymptotic Safety, non-perturbative fRG calculations demonstrate that quantum spacetime undergoes dimensional reduction at short distances¹. The spectral dimension d_s , defined via the return probability of a diffusion process governed by the heat kernel $K(x, x'; s)$, transitions continuously from $d_s = 4$ in the infrared (IR) regime to $d_s = 2$ at the ultraviolet (UV) fixed point¹:

$$d_s(k) = -2 \frac{d \ln P(k)}{d \ln k}$$

This dimensional reduction indicates that quantum spacetime exhibits self-similar, fractal properties near the Planck scale¹.

In Invariant Set Theory, the cosmological state space I_U is inherently fractal, with a Hausdorff dimension $D_H < D_{\text{Euclidean}}$ determined by the order p of the underlying p -adic structure³. Connecting these frameworks requires establishing a relationship between the energy scale k in the fRG flow and the p -adic resolution parameter $p(k)$ ¹. In this picture, the physical scale-dependence of spacetime metrics in Asymptotic Safety is interpreted as the physical spacetime manifestation of an underlying p -adic fractal manifold in state space³.

Triadic Domain: BM + IST + AS

The triadic integration of Bohmian Mechanics, Invariant Set Theory, and Asymptotic Safety represents a **Verified Research Gap**. Combining these three pillars yields a formulation where ontic particle and field trajectories evolve on a scale-dependent, non-computable fractal manifold $I_U(k)$ ². In this framework, the guidance equations governing particle paths are driven by effective average actions Γ_k operating at a non-trivial UV fixed point, resolving non-locality by restricting state space to a local p -adic invariant geometry¹.

Technical Gap & Barrier Mapping

Synthesizing Bohmian Mechanics, Invariant Set Theory, and Asymptotic Safety into a unified framework requires resolving several fundamental mathematical friction points.

Continuous Configuration Space vs. Discrete p-Adic Metric Topology

Standard Bohmian mechanics relies on smooth differential manifolds, defining trajectory velocity fields via spatial derivatives $\nabla_i S(q, t)$ on continuous Euclidean spaces $\mathbb{R}^{3N}$ ². Conversely, Invariant Set Theory models physical reality on Cantor sets homeomorphic to the

space of p -adic integers $\mathbb{Z}_p$, where standard spatial derivatives do not exist³. Bridging this gap requires constructing a scale-dependent mapping that translates discrete p -adic state transitions into smooth differential flows in the singular limit $p \rightarrow \infty$ ³. This is achieved using p -adic analysis tools, such as the Vladimirov operator (a fractional derivative operator on p -adic spaces), which aligns p -adic diffusion processes with continuous Schrödinger and Hamilton-Jacobi dynamics in the continuum limit³.

Relativistic Covariance and Background Independence vs. Preferred Foliation

Bohmian mechanics' explicit non-locality requires a preferred spatial foliation Σ_t to define instantaneous multi-particle guidance equations, conflicting with relativistic covariance². Meanwhile, Asymptotic Safety emphasizes background independence by employing a split-symmetry field decomposition $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$ in the fRG framework to preserve diffeomorphism invariance¹. Incorporating Invariant Set Theory resolves this conflict³. Because IST reproduces quantum correlations locally through non-computable state-space geometries rather than non-local forces, it eliminates the requirement for instantaneous configuration-space action³. This allows particle and field trajectories to be guided locally by scale-dependent phase gradients $\partial_\mu S_k(x)$ on a four-dimensional pseudo-Riemannian manifold, preserving relativistic covariance and background independence³.

Scheme Dependence in fRG Cutoffs vs. Ontic Field Trajectories

The fRG effective average action $\Gamma_k[\phi]$ relies on an artificial infrared regulator kernel $R_k(p^2)$ that suppresses modes with momenta $p^2 < k^2$ ¹. While physical observables must be scheme-independent at the physical limit $k \rightarrow 0$, intermediate scale-dependent quantities—including running couplings and derived quantum potentials $Q_k(x)$ —depend on the choice of R_k ¹. If Bohmian trajectories are derived from a scale-dependent potential $Q_k(x)$, the paths could vary depending on the chosen regulator R_k . Resolving this scheme dependence requires proving that physical particle paths derived from the scale-dependent Hamilton-Jacobi equation are invariant under transformations of R_k , particularly near the UV fixed point g_* , where scale invariance stabilizes the effective field trajectories¹.

Koopman-von Neumann Spectral Chaos and Non-Computability

The Koopman-von Neumann (KvN) formulation maps classical deterministic phase-space trajectories into a Hilbert space framework using unitary operators generated by the Liouville operator $L = -i \left(\frac{p}{m} \nabla_x - \nabla V \cdot \nabla_p \right)$ ¹⁴. When applied to non-linear chaotic systems evolving on fractal Cantor sets—such as I_U in IST—the KvN spectrum transitions from discrete eigenvalues to a continuous spectrum exhibiting spectral chaos³. This continuous spectrum complicates the extraction of smooth quantum wave-functionals $\Psi[g_{\mu\nu}, \phi]$ from deeper deterministic state-space dynamics³. Addressing this barrier requires applying coarse-graining techniques grounded in information geometry, using minimum Fisher information metrics to smooth chaotic KvN operators into stable Hilbert space representations¹³.

Feature Extraction Matrix for Framework Optimization (Version 6.3)

To construct Framework Version 6.3, explicit mathematical equations, theorems, and structural tools are extracted from adjacent physics literature to enhance the framework's coherence, cohesiveness, robustness, and compellingness.

Mathematical Tool / Theorem	Source Reference / Context	Core Mathematical Formulation / Equation	Target Structural Benefit to Version 6.3
Exact Uncertainty Principle & Information Geometry	Hall & Reginatto (2002) ¹⁶	$H =$	Coherence: Derives the Bohmian quantum potential directly from minimum Fisher information, linking statistical ensemble geometry to mechanical dynamics.
Weyl Geometry Identification of	Santamato (1984), Castro (2020) ⁷	$Q = \frac{\hbar^2}{2m} R_W, \quad A_\mu$	Cohesiveness: Geometrizes the Bohmian quantum

Quantum Potential			potential as the scalar curvature R_W of a Weyl connection, eliminating ad-hoc potential insertions.
Wetterich Functional RG Equation	Wetterich (1993), Reuter (1998) ¹	$\partial_k \Gamma_k = \frac{1}{16\pi^2} \text{Tr} \left[\dot{\Gamma}^{(2)} \circ \Delta_k^{-1} \right]$	Robustness: Drives the scale-dependent evolution of field couplings toward a non-trivial UV fixed point g_* , preventing Landau poles in the pilot wave.
p -Adic State-Space Metric & Cantor Integration	Palmer (2016, 2020) ³	$d_p(x, y) = p^{-k}$,	Robustness: Provides immunity to Bell, PBR, and Kochen-Specker theorems by restricting counterfactual states to non-computable orbits off I_U .
Quantum Raychaudhuri Equation (QRE)	Das (2014), Ali & Khalil (2015) ⁶	$\frac{d\theta}{d\tau} = -\frac{1}{3}\theta^2 - R_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$	Compellingness: Replaces gravitational spacetime singularities with smooth quantum bounces across cosmic evolution without requiring metric quantization.

Koopman-von Neumann Operator Formalism	Koopman (1931), Bondar et al. (2012) ¹⁴	$i \frac{\partial \psi_{\text{KvN}}}{\partial t} =$	Coherence: Maps microscopic deterministic classical trajectories directly into a unitary Hilbert space architecture.
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Theoretical Synthesis & Actionable Roadmap for Framework Version 6.3

The synthesis of Bohmian Mechanics, Invariant Set Theory, and Asymptotic Safety into Framework Version 6.3 establishes a multi-scale, micro-deterministic model of quantum gravity. The architecture operates across three integrated levels:

1. Micro-Deterministic Layer (Invariant Set Theory Geometry)

The universal state space is constrained to a non-computable fractal invariant set I_U in cosmological state space. The p -adic metric $d_p(x, y) = p^{-k}$ defines closeness between cosmological aeon histories³. This local determinism evades Bell inequalities without invoking spooky action-at-a-distance, as counterfactual measurement choices correspond to states off I_U and are physically unrepresentable³.

2. Scale-Dependent Hydrodynamic Layer (Bohmian / Weyl Geometry)

Ontic field trajectories are guided locally by scale-dependent velocity fields $v_k^\mu = \frac{1}{m} g_k^{\mu\nu} \partial_\nu S_k$, where S_k is the phase of the effective average wave-functional Ψ_k ¹. The quantum potential Q_k is geometrized via the scalar curvature $R_W(k)$ of a Weyl connection, uniting quantum potential dynamics with spacetime curvature⁷. The Fisher information metric guarantees self-consistent ensemble equilibrium $\rho_k = |\Psi_k|^2$ along the trajectory flow¹⁶.

3. Non-Perturbative UV Renormalization Layer (Asymptotic Safety)

The scale evolution of the average action Γ_k is governed by the Wetterich equation¹. As energy approaches the Planck scale ($k \rightarrow \infty$), gravitational and field couplings flow toward

a non-trivial UV fixed point (g_*, λ_*) , eliminating ultraviolet divergences and resolving spacetime singularities into smooth quantum bounces¹. The reduction of spacetime spectral dimension ($d_s \rightarrow 2$) at short distances directly reflects the underlying p -adic fractal structure of I_U ¹.

Mathematical Architecture of Version 6.3

Framework Version 6.3 models the universe U as a deterministic dynamical system evolving on an invariant manifold $I_U \subset \Omega$. The state space is governed by a p -adic metric $d_p(x, y) = p^{-k}$, where the order parameter $p(k)$ scales with energy k according to the non-perturbative fRG flow¹:

$$p(k) = \left\lfloor \left(\frac{k}{\Lambda_{\text{Planck}}} \right)^{-\alpha} \right\rfloor + 1$$

As $k \rightarrow 0$ (the infrared limit), $p \rightarrow \infty$, and the p -adic state space smoothly approaches a continuous complex Hilbert space $\mathcal{H}$ ³. At the ultraviolet limit ($k \rightarrow \infty$), p remains finite, enforcing the non-computable p -adic fractal geometry of I_U near the Planck scale³.

Field trajectories $\phi_k(x)$ on I_U are governed by a scale-dependent guidance equation derived from the phase $S_k[\phi]$ of the effective wave-functional $\Psi_k[\phi] = R_k[\phi] e^{iS_k[\phi]/\hbar}$ ¹:

$$\frac{\partial \phi_k(x)}{\partial \tau} = \frac{\delta S_k[\phi]}{\delta \phi_k(x)} \Big|_{\phi=\phi_k}$$

The wave-functional $\Psi_k[\phi]$ evolves according to the scale-dependent functional Hamilton-Jacobi equation⁷:

$$\frac{\partial S_k}{\partial \tau} + \int d^3x \left[\frac{1}{2\sqrt{g}} \left(\frac{\delta S_k}{\delta \phi} \right)^2 + V_k[\phi] + Q_k[\phi] \right] = 0$$

The quantum potential functional $Q_k[\phi]$ is geometrized via the Weyl scalar curvature

$R_W(k)$ 7.

$$Q_k[\phi] = -\frac{\hbar^2}{2} \frac{\square_k R_k[\phi]}{R_k[\phi]} = \frac{\hbar^2}{2} R_W(k)$$

The scale evolution of $V_k[\phi]$ and $Q_k[\phi]$ is regulated by the Wetterich functional renormalization group equation¹:

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)} + R_k \right)^{-1} \partial_k R_k \right]$$

This ensures that as $k \rightarrow \infty$, the theory approaches a non-trivial UV fixed point g_* , preventing Landau poles and UV divergences¹.

Strategic Roadmap for Framework Optimization

1. **Formalize the $p(k)$ Mapping Function:** Develop the analytical derivation connecting the fRG spectral dimension $d_s(k)$ to the p -adic state-space order parameter $p(k)$, demonstrating that UV spectral dimension reduction ($d_s \rightarrow 2$) arises from the p -adic fractal structure of I_U ¹.
2. **Prove Trajectory Scheme Invariance:** Apply information geometry to establish that particle and field trajectories $\phi_k(x)$ derived from $S_k[\phi]$ are invariant under transformations of the fRG regulator $R_k(p^2)$ near the UV fixed point g_* ¹.
3. **Derive Cosmological Bounce Conditions:** Solve the Quantum Raychaudhuri Equation (QRE) using the scale-dependent Bohmian potential $Q_k(x)$ at g_* , calculating singularity-free cosmological bounce solutions for the early universe¹.
4. **Establish Observational Benchmarks:** Compute modified primordial power spectra and gravitational-wave polarization signatures derived from the scale-dependent guidance equations, comparing the empirical predictions of Version 6.3 against standard inflationary cosmological models¹.

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